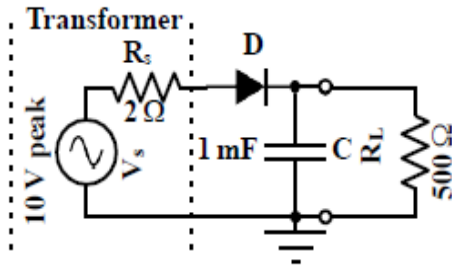


Tutorial 6 _ Answer key

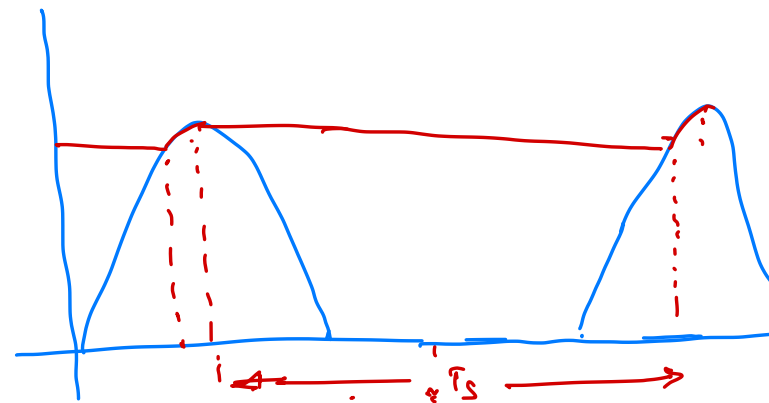
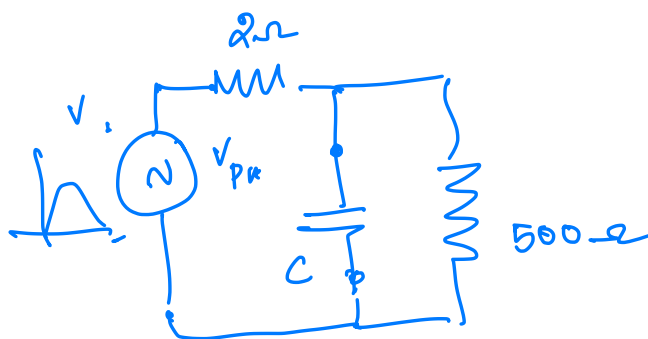
Q-1 A step down transformer generates an AC voltage of 10 V peak from 220 V, 50 Hz AC mains. The resistance of secondary winding is $2\ \Omega$. A diode rectifies the stepped down AC voltage and the half wave rectified AC is smoothed using a 1 mF capacitor. The load can be represented by a resistor of $500\ \Omega$. The resulting equivalent circuit is shown below. We'll assume the diode to be ideal in this problem.



During the positive half cycle, the diode conducts only when the AC voltage exceeds the voltage stored on the capacitor. Once the AC voltage drops below the capacitor voltage, the diode becomes an open circuit. In this state, the capacitor discharges through the load resistor.

- What is the time constant during the charge phase? (Replace the diode by a short circuit and the AC voltage source along with R_s and R_L to its Thevenin equivalent).
- What is the time constant during discharge? (Diode is replaced by an open circuit).
- Since the discharge time constant is much longer, we make the simplifying assumption that the discharge duration is nearly the whole AC cycle of 20 ms. Assume that the capacitor charges up to the Thevenin voltage during the short charging cycle. To what voltage will the capacitor discharge before the charge cycle begins again?
- What is the peak to peak ripple voltage?

The o/p waveform is shown,
ckt for charging

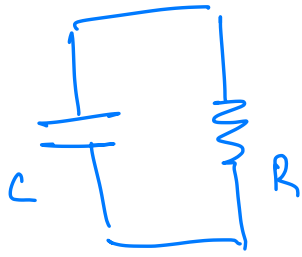


$$R_{TH} \text{ at capacitor} = 2 \parallel 500 = \frac{1000}{502} = 1.992\ \Omega$$

$$V_{TH} (pk) = 10 \times \frac{500}{502} = 9.96\text{ V}$$

$$\text{Charging Time Constant} = R_C = 1.992 \times 1 \times 10^{-3} = 1.992\text{ ms}$$

Time constant during discharge



$$= RC = 1 \times 10^{-3} \times 500$$

$$= 0.5 \text{ sec} = 500 \text{ ms}$$

Ckt during discharge
(Diode is open)

$$\text{AC cycle time} = T_s = \frac{1}{f} = 20 \text{ ms},$$

discharging $> T_s$

charging time $< T_s$

While discharging Max voltage $V_0 = 9.96 \text{ V}$

$$\text{Min voltage} = V_0 e^{-\frac{20}{500}}$$

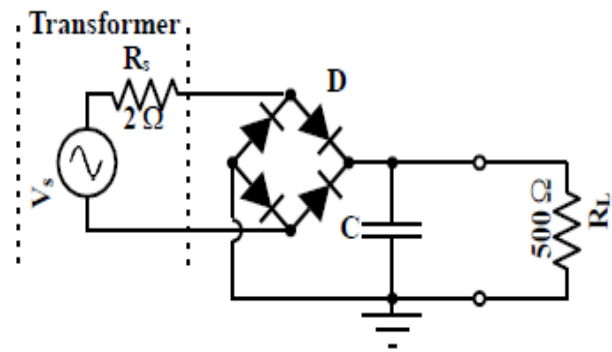
$$\approx 9.57 \text{ Volts}$$

(d) Peak ripple voltage

$$= V_{0_{\text{max}}} - V_{0_{\text{min}}} = 9.96 - 9.57$$

$$= 390 \text{ mV}$$

Q-2 You are tasked with designing a Full-Wave Bridge Rectifier circuit with a capacitor filter to supply a steady 10 V DC to a 500 Ω load. The circuit is powered by a standard 50 Hz AC transformer. The output ripple voltage $V_{r(pp)}$ must not exceed 0.2 V. Assume standard silicon diodes with a forward drop of 0.7 V.



- Calculate the minimum capacitance (C) required to meet the ripple specification.
- Determine the required Peak Secondary Voltage (V_m) and the corresponding RMS Voltage rating required from the transformer to maintain the 10 V DC output.

The Load Waveform is shown

using the c/p discharge equation

$$v(t) = V_{max} e^{-t/RC}$$

$$V_{max} = 10 \pm 0.1 \Rightarrow 10.1$$

$$V_{min} = 9.9$$

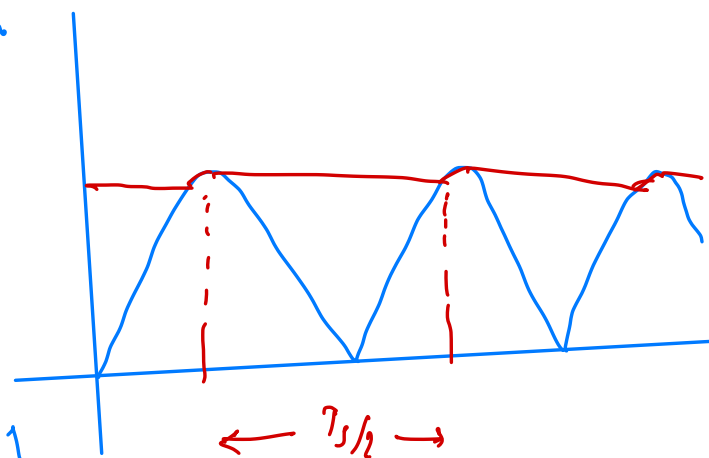
$$V_o(t)_{min} = V_o(t)_{max} e^{-t/RC}$$

$$e^{-t/RC} = \frac{9.9}{10.1} = 0.9802$$

$$\frac{t}{RC} = 0.02$$

$$\therefore RC = \frac{10 \times 10^{-3}}{0.2} \approx 0.5 \text{ sec.}$$

$$\therefore C = \frac{0.5}{500} = 1 \text{ mF.}$$



Peak Secondary voltage = $10 + 0.1 + I_{dc} \times 2 + 0.7 \times 2$
(Including voltage drop of diodes)

I_{dc} : Current through 2 Ohm resistor at $V_s = 10V = 10/500 = 0.02 A$

$$= 11.54 \text{ volt}$$

Transformer secondary voltage

$$= \sqrt{2} V_s = 11.54$$

$$V_s = 8.1$$

(rms)

Q-3 You are designing a 5.0 V regulated power supply and evaluating two possible circuit configurations, A) A single Zener Diode in Reverse Breakdown, B) A series string of 7 standard Silicon Diodes in Forward Bias.

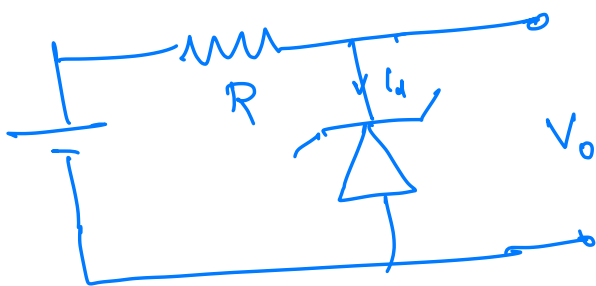
You decide to compare the two designs using the manufacturer's datasheet values below.

A) Zener Diode (Reverse Biased): Voltage at low current (10 mA): $V_Z = 4.95$ V and Voltage at high current (100 mA): $V_Z = 5.05$ V,

B) Standard Diode (Forward Biased): Voltage at low current (10 mA): $V_F = 0.70$ V and Voltage at high current (100 mA): $V_F = 0.85$ V.

- Calculate the change in total output voltage (i.e. voltage regulation) for both designs when the current increases from 10 mA to 100 mA.
- Based on your calculation in part (a), determine which design is better for a voltage regulator. Explain your choice by comparing the Dynamic Resistance (slope of the V-I curve) of the two designs.

Zener diode Configuration



Zener Specs

$$V_Z = 4.95 \text{ V} \quad I_d = 10 \text{ mA}$$

$$V_Z = 5.05 \text{ V} \quad I_d = 100 \text{ mA}$$

$$R_Z = \frac{\Delta V}{\Delta I} = \frac{0.1}{90 \times 10^{-3}}$$

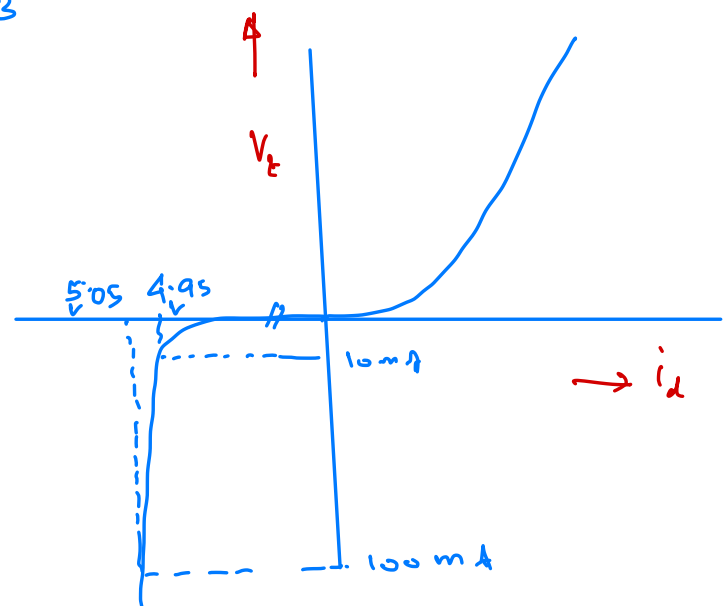
$$= 1.11 \, \Omega$$

Voltage drop

for a 90 mA current

$$Z_{\text{ener}} = 1.11 \times 90 \text{ mA}$$

$$\Delta V_o = 0.1 \text{ Volts}$$



Standard diode (FWD biased) 7 in series

$$V_o \underset{\text{min } I_L}{=} 0.7 \times 7 = 4.9 \text{ Volts}$$

$$V_o \underset{\text{for Max } I_o}{\text{max}} = 0.85 \times 7 = 5.95$$

$$\begin{aligned} \Delta V_o &= 5.95 - 4.9 \\ &= 0.95 \text{ Volts} \end{aligned}$$

Voltage regulation in the case
of Zener diode

$$= \frac{\Delta V_o}{V_o} \times 100$$

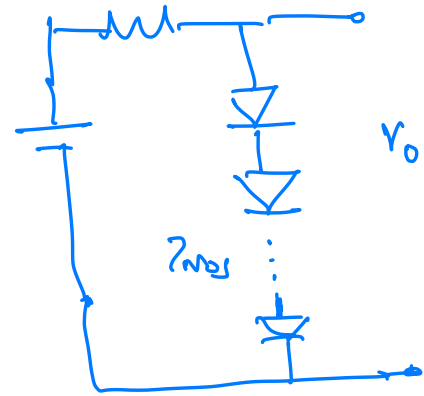
$$\text{Let } V_o = 4.95 \text{ V}$$

$$\% \text{ regulation} = \frac{0.1}{4.95} \times 100 = 2.02 \%$$

% Voltage regulation in the case of Standard diode

$$= 100 \times \frac{0.95}{4.9} = 19.38 \%$$

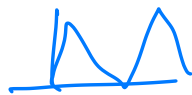
Clearly the Zener diode based voltage
regulator is better



Q-4 You are designing an interface card for a vibration sensor which outputs an AC sine wave that swings between -5 V and $+5\text{ V}$. The microcontroller has an analog input that can only safely handle positive voltages (0 V to 10 V). If the signal goes negative, the chip will be damaged. To fix this, you design a interface circuit (using a capacitor and a diode) between the sensor and the microcontroller. Given: Sensor Frequency (f): 1 kHz , Microcontroller Input Resistance (R): $10\text{ k}\Omega$. Assume the Diode is Ideal (acts as a perfect switch with 0 V drop when ON).

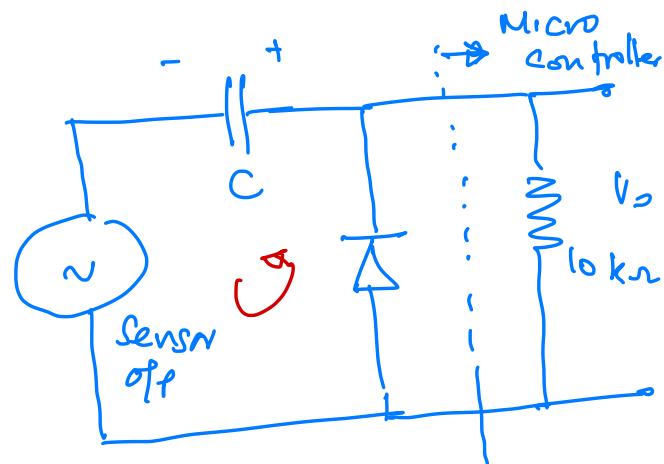
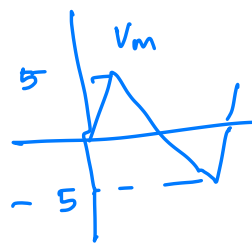
Show the design of the interface circuit, and its analysis which results in desired output. Discuss the choice of the value of the capacitor.

What we need is a +ve clamper which maintains only a positive voltage output.



Circuit is shown

Capacitor initially uncharged, will get charged during



-ve cycle to $2V_m$ (in Half cycle from 180° to 360°)

once charged to max $2V_m$ and and addition

of Triangular wave with $\pm 5\text{ V}$ will make the

voltage across load vary from $2V_m$ to 0 V to

$$(or) +V_m \pm (V_m) = 2V_m \text{ to } 0 \text{ volts}$$

Thus the microcontroller voltage is from 0 to $2V_m$

The average o/p voltage is V_m .

Since the charging time constant R_C
when the diode is conducting is zero
the cycle time of the triangular wave
is 1 ms.

The Time Constant of the Capacitor
charging = $C R_{d(on)}$

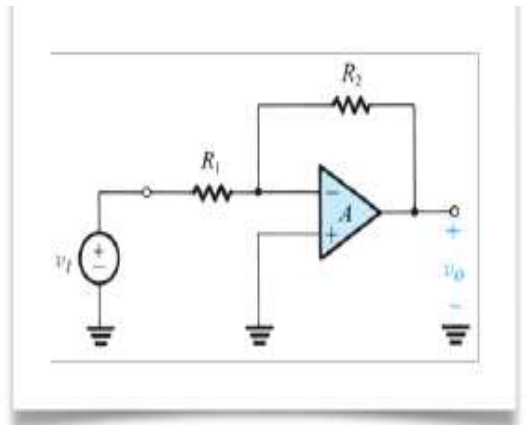
C should be chosen such that

$$R_d C \ll \ll 1 \text{ ms}$$

Q-5 In your design of a precision amplifier circuit for a medical device, you have chosen an inverting amplifier configuration with resistor values $R_1 = 1 \text{ k}\Omega$ and $R_2 = 10 \text{ k}\Omega$.

Your purchasing manager has offered you two Op-Amp chips to choose from: A) a Gold standard chip with an Open-Loop Gain of $A = 100,000$ but costs 1000 Rupees and, B) a Budget chip with an Open-Loop Gain of $A = 1000$, but cost 30 Rupees.

To make your decision, you are advised to



- First calculate the ideal closed-loop gain assuming $A = \infty$.
- Then, determine the actual closed-loop gain for both the Gold and Budget chips.
- Finally, calculate the percentage error of the Budget chip compared to the ideal target and determine if the cheaper chip is sufficient for a design requiring a 1% tolerance.

The op Amp is in inverting mode,

The gain of the inverting mode

(a) Gain of ideal op
 Gain is given by $A_v = \frac{-R_f}{R_{in} + \frac{R_f + R_{in}}{A_{OL}}}$

When $A_{OL} \rightarrow \infty$, $R_f = 10 \text{ k}$
 $R_i = 1 \text{ k}$

$$A_v = - \frac{10}{1 + \frac{11}{\infty}} = -10$$

(b) for Gold standard gain = 100000

$$A_v = \frac{-10}{1 + \frac{11}{100000}} = - \frac{10}{1 + \frac{11}{100000}}$$

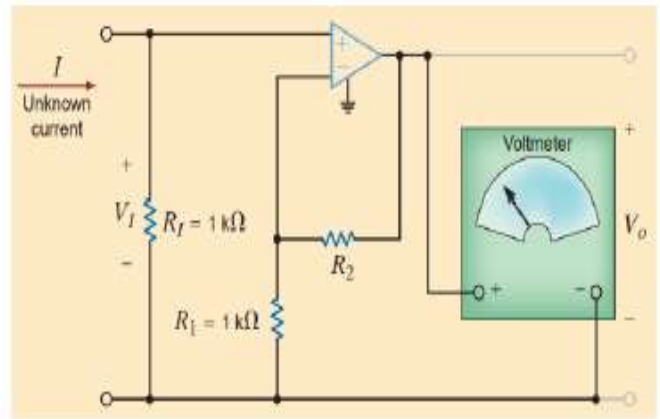
$$= -9.9989 \quad (0.01\%) \text{ error Tol}$$

for Budget chip $A_v = \frac{-10}{1 + \frac{11}{1000}}$
 $= -9.891 \quad (1.09\%) \text{ error Tol}$

% error of the budget check is $> 1\%$

The Gold standard meets the requirement

Q-6 You are given a circuit to design an electronic ammeter circuit that measures an unknown current, I , by first converting it into a voltage and then amplifying it for a display. The input current flows through a shunt resistor, $R_I = 1 \text{ k}\Omega$, to ground, creating a small sensing voltage (V_I). This voltage is then fed into a non-inverting Op-Amp configuration defined by an input resistor $R_1 = 1 \text{ k}\Omega$ and a feedback resistor R_2 .



The goal is to calibrate the meter so that the output voltage (V_{out}) is exactly 10 V for every 1 mA of input current. Determine the value of the feedback resistor, R_2 , required to achieve this specific sensitivity.

Assuming Ideal Op Amp

The gain of the non-

Inverting op amp

$$= 1 + \frac{R_f}{R_{in}}$$

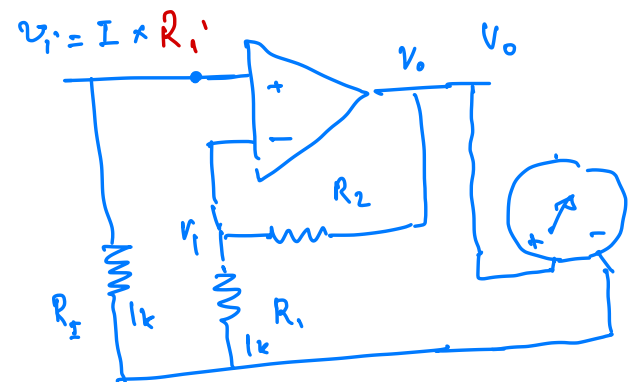
$$\frac{V_o}{V_{in}} = 1 + \frac{R_2}{R_1}$$

$$V_o = \left(1 + \frac{R_2}{1}\right) V_{in} = \left(1 + \frac{R_2}{R_1}\right) \cdot 1$$

$$10 = \left(1 + \frac{R_2}{1}\right) 1$$

$$\therefore R_2 = 10 - 1 = 9 \text{ k}\Omega$$

$$R_2 = 9 \text{ k}\Omega$$



$$V_{in} = 1 \text{ mA} \times 1 \text{ k}\Omega = 1 \text{ V}$$